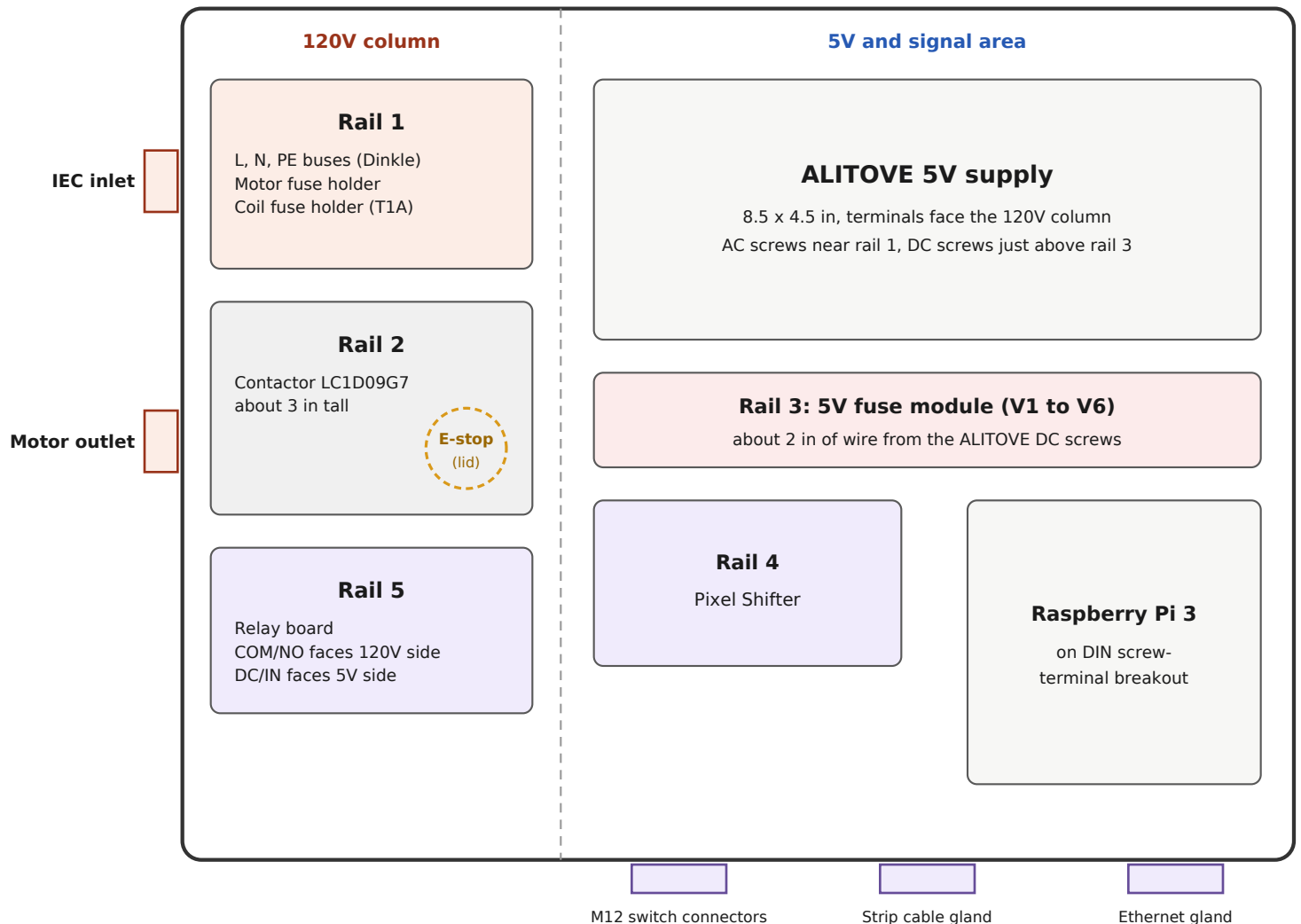


What's in this reference

- 1 Box layout: where every component sits (this page)
- 2 120V wiring: inlet, buses, ALITOVE, contactor, e-stop, relay, motor outlet
- 3 Mains rail 1 close-up: every Dinkle block, clamp and short jumper wire
- 4 5V power: ALITOVE DC side, fuse module channels, and each 5V load
- 5 Signal wiring: Pi GPIO to switches, relay, Pixel Shifter and the strip data
- 6 Strip-end breakout box: capacitor and 330 ohm resistor before the first pixel

Box interior, about 12.5 in x 9 in (rough layout, not to scale)



How to read these pages

- | | |
|------------------------------------|--------------------------------------|
| — Hot, 120V (black 16 AWG) | — +5V |
| — Neutral, 120V (white 16 AWG) | — 5V ground (drawn blue for clarity) |
| — Earth (green 16 AWG) | — Signal wire (3.3V / 5V logic) |
| — Short jumper wire between blocks | — - - - Optional route |

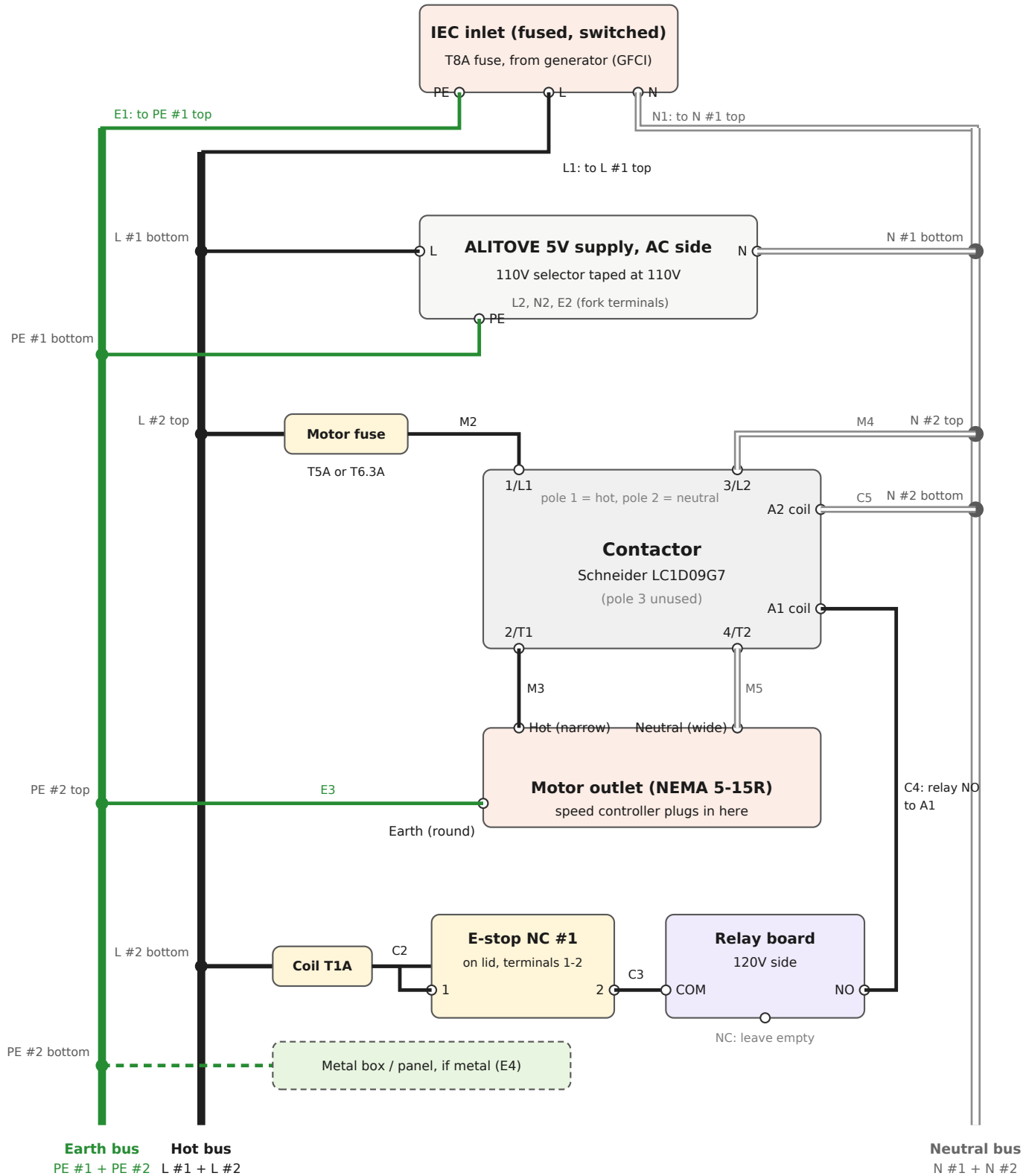
A dot on a wire is a real connection. Wires that cross without a dot do not touch.

Labels like L1, M2, C3 in brackets are the wire names from the guide's wire list.

Mains work is done unplugged. Wait a minute after unplugging before touching the AC side.

120V wiring (mains side)

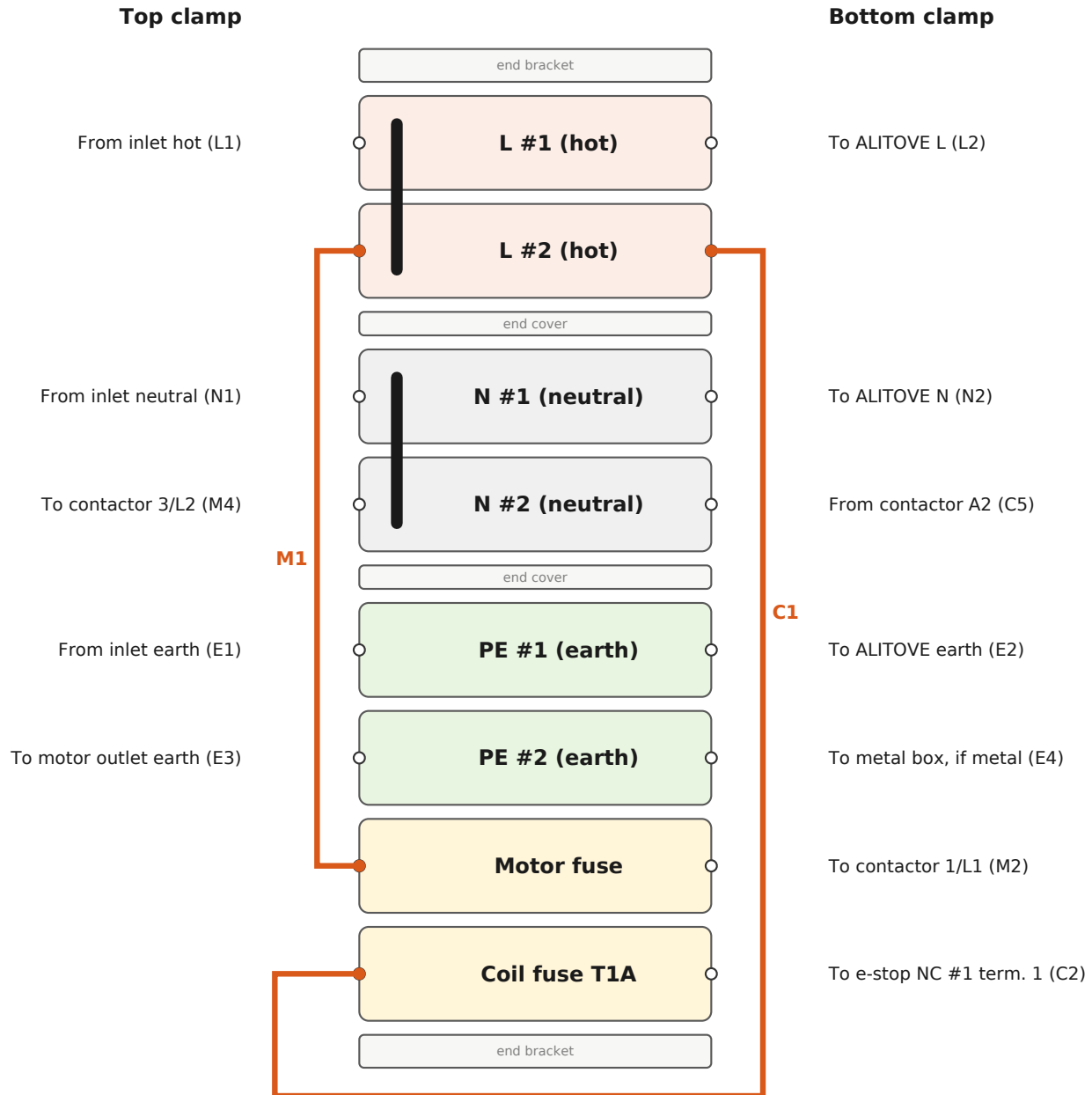
Every 120V component and the wire between each pair. The three buses are the Dinkle blocks on rail 1 (close-up on page 3).



The e-stop and relay only carry the coil's small current. The contactor carries the motor current. Earth is never switched or fused: inlet to earth bus to outlet. Hot AND neutral both pass through the contactor. Contactor terminal positions are simplified here; go by the labels printed on the contactor. Wires that cross the buses without a dot do not connect to them.

Mains rail 1 close-up (Dinkle blocks)

Turned on its side to fit the page. On the real rail the blocks sit side by side; left = top clamp, right = bottom clamp.



Jumper wire M1 (orange): from L #2's top clamp to the motor fuse holder's IN clamp. Black 16 AWG, ferrule each end.

Jumper wire C1 (orange): from L #2's bottom clamp, around the rail end, to the coil fuse holder's IN clamp.

Thick black bars = metal comb jumpers (DSS2.5N) pressed into the centre slots. They join L #1 to L #2 and N #1 to N #2.

PE blocks have no comb: they connect through the metal rail. The inlet earth still lands directly on PE #1.

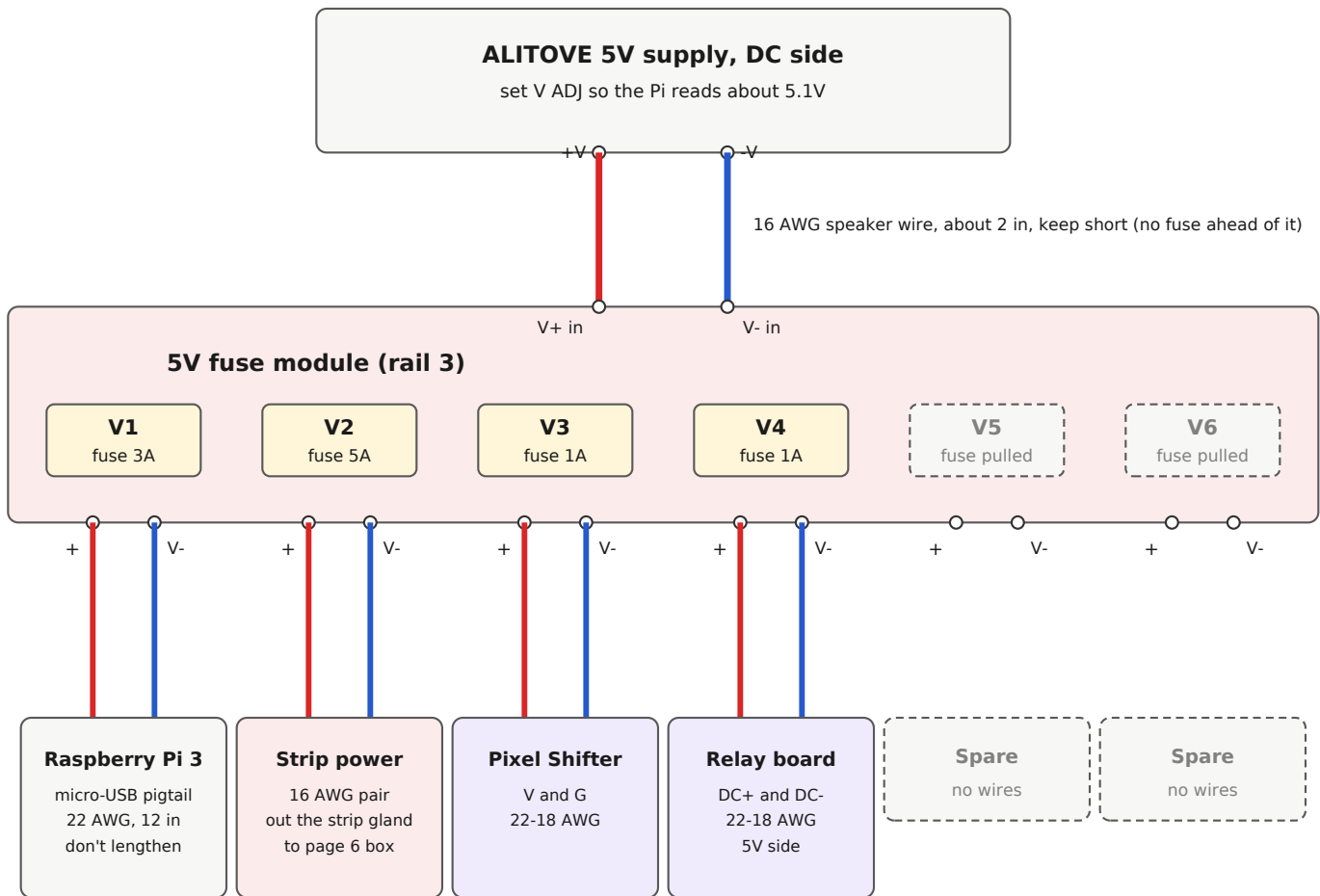
Orange lines = the two short wires you cut and install. Every clamp holds exactly one wire.

Meter check before wiring: L #1-L #2 beep, N #1-N #2 beep, any L to any N silent, PE #1-PE #2-rail beep.

Landing a wire: strip, crimp a ferrule, push it into the clamp's metal box, tighten firmly, tug hard.

5V power distribution

ALITOVE DC output to the fuse module, then one fused channel per 5V load: + from V_n, ground from the V- beside it.



Before the first plug-in (unplugged, meter on continuity)

Module input V+ to V-: no beep. Each output V_n to its V-: no beep.

Then plug in, switch on, and set V ADJ so the Pi's input reads about 5.1V.

Pass: the Pi boots and the dashboard shows a green PWR OK pill.

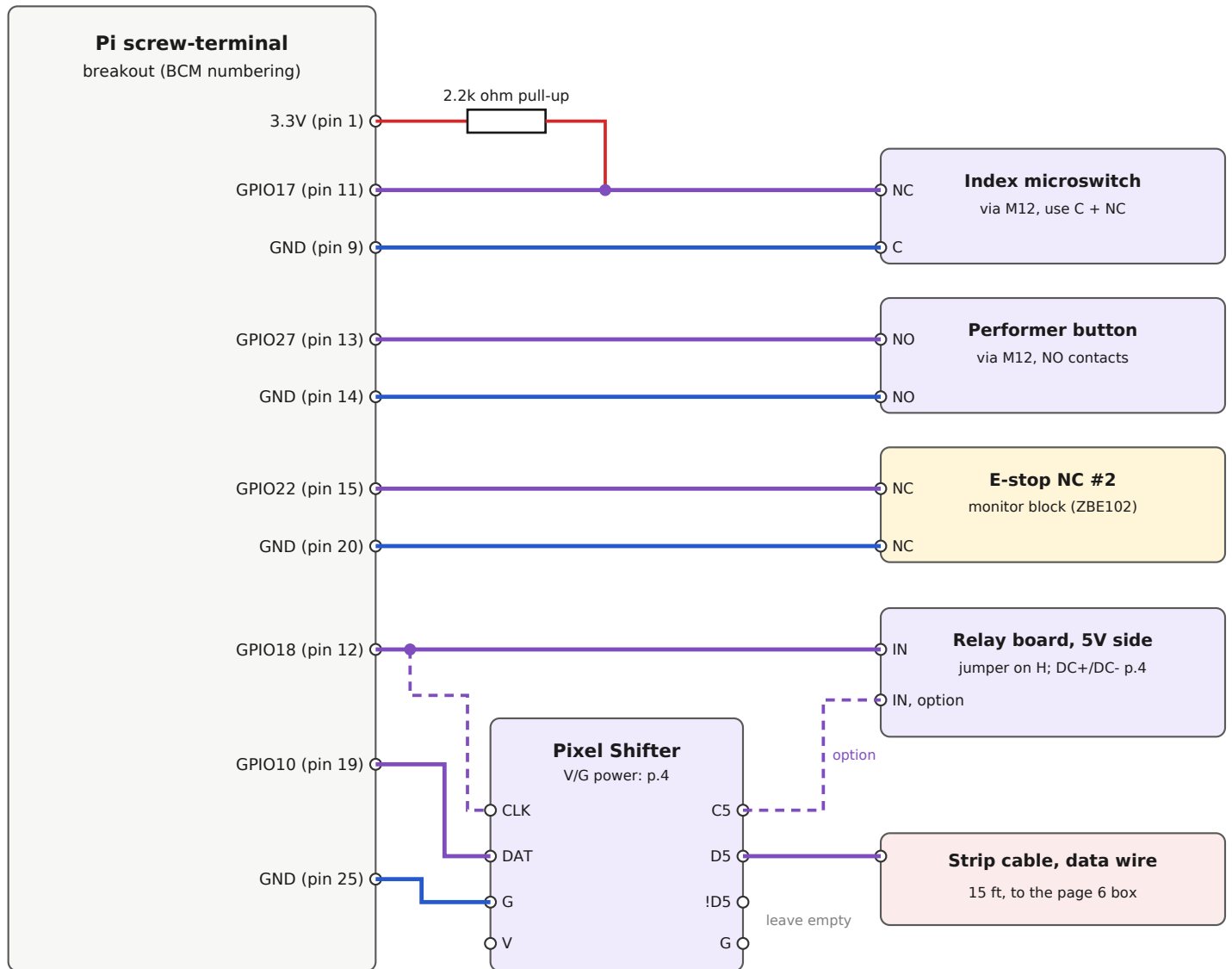
Why each load has its own fuse

A short on the strip (15 ft away, on a moving float) blows only V2. The Pi keeps running and can show what happened.

V3 and V4 are 1A because the Pixel Shifter and relay board draw only milliamps; a fault can't overheat their thin wires.

The Pi, Pixel Shifter, relay board and strip all share this ground, which the data and relay signals need.

Every GPIO wire, what it connects to, and the Pixel Shifter's two lanes. All switches connect a GPIO pin to GND only.



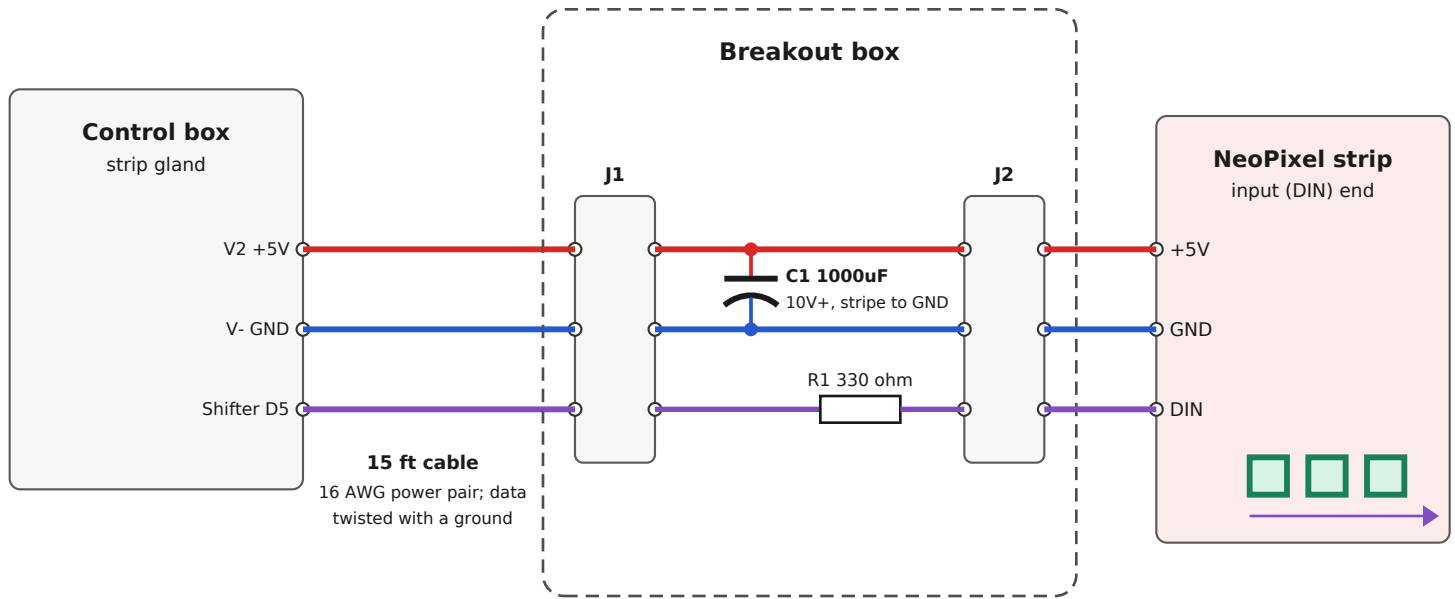
Relay IN is ONE terminal, shown twice. Use one route only: the guide wires GPIO18 straight to IN. If the relay won't trigger reliably from 3.3V, use the dashed lane instead: GPIO18 to Shifter CLK, C5 to IN, direct wire removed. Index switch is wired through NC so a cut wire reads as 'at index' and stops the motor. The 2.2k pull-up gives the heavy-duty microswitch enough current to switch reliably (goes at the box end). Every GND terminal on the breakout is the same connection; the pin numbers shown are just the nearest ones. Twist each switch's signal and ground together and keep switch cables away from the motor and mains wires.

Bench tests (service stopped)

```
.venv/bin/parade-hwcheck inputs      watch every switch change state
.venv/bin/parade-hwcheck relay --seconds 3  relay clicks for 3 s, COM-NO beeps only then
.venv/bin/parade-hwcheck pixels test    red, green, blue, white, then a chase
```

Strip-end breakout box

A small sealed box at the end of the 15 ft run, just before the first pixel. Holds the capacitor and the 330 ohm data resistor.



Parts

J1, J2: 3-way screw terminal blocks (5.08 mm pitch takes 16 AWG), or a short DIN rail with three block pairs.

C1: 1000uF electrolytic, 10V or higher (16V common). The striped leg is negative and goes to GND.

R1: 330 ohm, 1/4 W, close to J2 so it sits as near the first pixel as possible.

Optional: 0.1uF ceramic right beside C1. Enclosure: small weatherproof box with two cable glands.

Building it

Solder on perfboard with J1 on one edge and J2 on the other. Run solid 16 AWG on the back for +5V and GND.

Label +5V / GND / DATA on the board and the lid. Keep the J2-to-DIN lead short (a foot or less).

Find the strip's input end: printed arrows point away from it and the pads read DIN, not DOUT.

Power and ground only reach the input end. Nothing returns from the far end; DOUT stays empty.

If the far end looks dim or yellow at full white, add power injection (+5V and GND only) at the far end,

or set brightness: 0.6 in the config.

Test before sealing

Run parade-hwcheck pixels test, then the full-white power stress test with the PWR pill green.

Only then close the box or apply adhesive-lined heat shrink.